

$G(3,1,1)$ Extension of Double Spacetime Torsional Mismatch: 10-Dimensional Generator Structure and Plane-Based Bracket Products

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Abstract

We embed the double spacetime structure into the projective geometric algebra $G(3, 1, 1)$ by adjoining an additional null vector e_4 to the biquaternion basis of $\text{Cl}(3, 1)$. The resulting torsional mismatch is generated by a ten-dimensional bivector space consisting of three hyperbolic generators (iI, iJ, iK) , three cyclic generators (I, J, K) , and four null generators $N_\mu := e_4 \wedge e_\mu$ ($\mu = 0, 1, 2, 3$) constructed from e_4 and the four usual-spacetime vector generators $\{j, kI, kJ, kK\}$. This ten-dimensional generator algebra reproduces the Lie-algebraic structure of the Poincaré algebra $\mathfrak{iso}(3, 1)$: the six hyperbolic-cyclic bivectors close into $\mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1)$, while the four null bivectors form an abelian translation ideal.

The quadratic invariant of the extended mismatch is built from a non-degenerate bilinear form that combines the Cartan–Killing form on the semisimple sector with the Minkowski metric on the translation sector, the latter compensating the well-known degeneracy of the Killing form on translations. We further establish a direct correspondence between the plane-based bracket products—the commutator and anticommutator of the geometric product on three-blades—and the property that the dual spacetime acts as the normal to three-blades. Because every null generator N_μ is a bivector that squares to zero and is reversal-odd ($\widetilde{N}_\mu = -N_\mu$), the extended motor $M = \exp(\Omega_{\text{biv}}^{(5)})$ is unitary, $M\widetilde{M} = 1$, thereby maintaining causal propagation. The construction yields a unified algebraic description of light degrees of freedom and torsional mismatch without invoking a background continuum.

1 Introduction

The double spacetime framework describes gravity and inertia as torsional mismatch between two intrinsically paired Minkowski structures carried by each massive particle. To incorporate translational degrees of freedom in a geometrically natural manner, we lift the construction into the projective geometric algebra $G(3, 1, 1)$ [1, 2, 3, 4], whose generating vector space is five-dimensional and whose full Clifford algebra is $2^5 = 32$ -dimensional. The use of a degenerate vector with vanishing square to encode translations as bivectors goes back to Selig’s treatment of rigid-body kinematics in conformal/projective Clifford algebras [11], and is now the standard PGA convention for both Euclidean and Minkowski signatures [2, 4]. The additional null vector e_4 anticommutes with the four $\text{Cl}(3, 1)$ vector generators $\{j, kI, kJ, kK\}$ [7, 6] and satisfies $e_4^2 = 0$. It moreover commutes with the biquaternionic pseudoscalar $i = jkIkJkK$ [10], so the duality map $X \mapsto Xi$ acts on the $\text{Cl}(3, 1)$ sub-algebra in the same way as in the parent theory while leaving the new null direction intact.

The mismatch bivectors partition into three distinct classes. The three hyperbolic generators iI, iJ, iK square to $+1$ and span the boost sector of $\mathfrak{so}(3, 1)$. The three cyclic generators I, J, K square to -1 and span the rotation sector of the dual copy of $\mathfrak{so}(3, 1)$. The four null generators $N_\mu := e_4 \wedge e_\mu$ ($\mu = 0, 1, 2, 3$), constructed from e_4 and the four $\text{Cl}(3, 1)$ vector generators, square to zero and form an abelian translation ideal. This yields a ten-dimensional bivector space whose Lie-algebraic structure matches the Poincaré algebra $\mathfrak{iso}(3, 1)$ [12, 13]. Translations therefore emerge directly from the null sector without auxiliary gauge fields, and the resulting extended mismatch bivector unifies the original torsional degrees of freedom with translational ones inside a single algebraic object.

A second novel feature appears when the plane geometry of $G(3, 1, 1)$ is examined. The plane-based bracket products—the commutator and anticommutator of the geometric product on three-blades—reproduce precisely the normal-mapping property that characterises the dual spacetime. This identification supplies a direct geometric interpretation of the dual sector as the normal bundle to three-blades,

thereby tightening the link between the algebraic structure and the projective geometry of the null direction.

Because each null bivector squares to zero *and* satisfies $N_\mu N_\nu = 0$ for every μ, ν , the exponential of the null part is a finite polynomial of degree one. Combined with the reversal-odd character $\widetilde{N}_\mu = -N_\mu$ of all bivectors, this guarantees the unitarity $M\widetilde{M} = 1$ of the extended motor and hence that the sandwich product preserves the degenerate metric of $G(3, 1, 1)$. Light-like propagation and torsional mismatch are therefore treated on an equal algebraic footing.

The paper is organised as follows. Section 2 defines the embedding of the double spacetime into $G(3, 1, 1)$ and classifies the ten generators. Section 3 constructs the extended mismatch bivector, gives a rigorous proof of the polynomial form of the null exponential, and introduces the non-degenerate bilinear form on the ten-dimensional generator space. Section 4 establishes the correspondence between the plane-based bracket products and the normal property of the dual spacetime. Section 5 verifies unitarity of the motor and outlines the resulting unified description of light and torsion. Conclusions are drawn in Section 6.

2 $G(3, 1, 1)$ Embedding and the Ten-Dimensional Generator Structure

2.1 Vector basis and the adjunction of e_4

The parent theory identifies the four $\text{Cl}(3, 1)$ vector generators with biquaternion elements

$$e_0 = j, \quad e_1 = kI, \quad e_2 = kJ, \quad e_3 = kK,$$

satisfying the Minkowski signature $(-, +, +, +)$:

$$e_0^2 = -1, \quad e_1^2 = e_2^2 = e_3^2 = +1, \quad \{e_\mu, e_\nu\} = 0 \ (\mu \neq \nu).$$

To incorporate translational degrees of freedom while preserving the torsional character of the double spacetime framework, we lift this structure to $G(3, 1, 1)$ by adjoining a single new vector generator e_4 characterised by

$$e_4^2 = 0, \quad \{e_4, e_\mu\} = 0 \ (\mu = 0, 1, 2, 3). \quad (1)$$

The full Clifford algebra $G(3, 1, 1)$ has dimension $2^5 = 32$; the original 16-dimensional biquaternion algebra is recovered as the sub-algebra generated by $\{e_0, e_1, e_2, e_3\}$.

2.2 Compatibility with the biquaternion duality map

The biquaternionic pseudoscalar of the parent theory is

$$i = e_0 e_1 e_2 e_3 = j(kI)(kJ)(kK),$$

and the duality map of the double spacetime acts as $X \mapsto Xi$. Using $\{e_4, e_\mu\} = 0$ repeatedly,

$$e_4 i = e_4 e_0 e_1 e_2 e_3 = (-1)^4 e_0 e_1 e_2 e_3 e_4 = i e_4,$$

so e_4 commutes with i . Consequently, the duality map $X \mapsto Xi$ operates on the $\text{Cl}(3, 1)$ sub-algebra exactly as in the parent paper, while e_4 is left intact as an independent null direction. This is the precise sense in which the duality is extended to $G(3, 1, 1)$ “while preserving the null character of e_4 ”.

2.3 Classification of the ten bivector generators

The bivector space of $G(3, 1, 1)$ has dimension $\binom{5}{2} = 10$. Combining the six $\text{Cl}(3, 1)$ bivectors with the four bivectors involving e_4 , the generators of the extended torsional mismatch fall into three distinct classes according to the signature of their squares:

- **Hyperbolic generators** (square to $+1$):

$$iI = e_0 e_1, \quad iJ = e_0 e_2, \quad iK = e_0 e_3.$$

These three elements span the boost sector of $\mathfrak{so}(3, 1)$.

- **Cyclic generators** (square to -1):

$$I = e_3 e_2, \quad J = e_1 e_3, \quad K = e_2 e_1.$$

These three elements span the rotation sector of the dual copy of $\mathfrak{so}(3, 1)$.

- **Null generators** (square to 0): four bivectors built from e_4 and the four $\text{Cl}(3, 1)$ vector generators,

$$\begin{aligned} N_0 &= e_4 \wedge e_0 = e_4 j, & N_1 &= e_4 \wedge e_1 = e_4 (kI), \\ N_2 &= e_4 \wedge e_2 = e_4 (kJ), & N_3 &= e_4 \wedge e_3 = e_4 (kK). \end{aligned}$$

Using (1), we have for any $\mu, \nu \in \{0, 1, 2, 3\}$,

$$N_\mu N_\nu = (e_4 e_\mu)(e_4 e_\nu) = -e_4^2 e_\mu e_\nu = 0, \quad (2)$$

so each N_μ squares to zero (the case $\mu = \nu$) and *any pairwise product* of null generators also vanishes. The four N_μ span an abelian translation ideal of the ten-dimensional algebra.

Collectively, these classes furnish a ten-dimensional bivector space:

$$\dim = 3 \text{ (hyperbolic)} + 3 \text{ (cyclic)} + 4 \text{ (null)} = 10,$$

matching the dimension of the Poincaré algebra $\mathfrak{iso}(3, 1)$. The semisimple part $\mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1)$ is generated by the six hyperbolic-cyclic bivectors, and the four null bivectors N_μ realise the translation generators P_μ .

2.4 Null sector and translations

The null sector directly encodes spacetime translations. Indeed, for any translation parameter a^μ ($\mu = 0, 1, 2, 3$), the combination $T(a) := \frac{1}{2} a^\mu N_\mu$ (Einstein summation implied) is a bivector built from the null sector alone. By (2), $T(a)^2 = 0$, so the exponentiation

$$\exp(T(a)) = 1 + \frac{1}{2} a^\mu N_\mu$$

truncates at first order. Consequently, the translational action appears as a pure shift without introducing additional curvature or higher-order terms; in particular, light-like translations correspond to the cone $\eta_{\mu\nu} a^\mu a^\nu = 0$ inside the four-dimensional parameter space.

2.5 Closure under duality

Since the duality map $X \mapsto Xi$ acts on the $\text{Cl}(3, 1)$ sub-algebra exactly as in the parent paper, it exchanges the boost generators $\{iI, iJ, iK\}$ with the rotation generators $\{I, J, K\}$ up to signs (this is the very statement that the usual and dual sectors are interchanged). Moreover, because e_4 commutes with i ,

$$N_\mu i = (e_4 e_\mu) i = e_4 (e_\mu i),$$

so duality maps the null sector into itself: each N_μ is sent to $e_4 \wedge (e_\mu i)$, which remains a bivector of the form $e_4 \wedge (\text{vector})$ and hence stays within the four-dimensional null subspace. Therefore the ten-dimensional generator space is closed under the duality operation, ensuring that the partition into hyperbolic, cyclic, and null components is preserved in both sectors. This closure property will be essential when we later examine the action of plane-based products on the dual spacetime.

In summary, the embedding into $G(3, 1, 1)$ organises the extended mismatch into a ten-dimensional bivector algebra isomorphic to the Poincaré algebra $\mathfrak{iso}(3, 1)$, in which translations emerge naturally from a four-dimensional null sector built from e_4 and the usual-spacetime vector generators.

3 Construction of the Extended Torsional Mismatch

We now construct the extended torsional mismatch bivector on the ten-dimensional generator space introduced in the previous section. Let $\Omega_{\text{biv}}^{(5)}$ denote the infinitesimal generator of the extended motor. We decompose it according to the three classes of generators:

$$\Omega_{\text{biv}}^{(5)} = \underbrace{\sum_{a=1}^3 \left(\frac{\alpha_a}{2} B_a^+ + \frac{\beta_a}{2} B_a^- \right)}_{\Omega_{\text{torsion}} : \text{hyperbolic} + \text{cyclic (6D)}} + \underbrace{\sum_{\mu=0}^3 \frac{\lambda^\mu}{2} N_\mu}_{\Omega_{\text{trans}} : \text{null sector (4D)}}, \quad (3)$$

where B_a^+ are the hyperbolic generators (iI, iJ, iK) , B_a^- are the cyclic generators (I, J, K) , and $N_\mu = e_4 \wedge e_\mu$ ($\mu = 0, 1, 2, 3$) are the four null bivectors constructed in Section 2.

The coefficients α_a and β_a parameterise the original torsional mismatch between the usual and dual sectors, while the four coefficients λ^μ encode the translational mismatch as a Lorentz four-vector. Light-like translations correspond to the cone $\eta_{\mu\nu} \lambda^\mu \lambda^\nu = 0$ inside the four-dimensional parameter space.

3.1 Rigorous truncation of the null exponential

Because the strong vanishing property (2) ensures $N_\mu N_\nu = 0$ for *all* pairs (μ, ν) —not merely for $\mu = \nu$ —we have

$$\left(\sum_{\mu=0}^3 \frac{\lambda^\mu}{2} N_\mu \right)^2 = \frac{1}{4} \sum_{\mu, \nu=0}^3 \lambda^\mu \lambda^\nu N_\mu N_\nu = 0, \quad (4)$$

and therefore the exponential of the null part truncates at first order:

$$\exp(\Omega_{\text{trans}}) = \exp\left(\sum_{\mu=0}^3 \frac{\lambda^\mu}{2} N_\mu\right) = 1 + \sum_{\mu=0}^3 \frac{\lambda^\mu}{2} N_\mu. \quad (5)$$

We emphasise that the validity of (5) requires *all* mixed products $N_\mu N_\nu$ to vanish, not only the diagonal squares $N_\mu^2 = 0$; the strong vanishing (2) is precisely what supplies this.

3.2 Motor factorisation

Because every N_μ commutes with itself (trivially) and because $N_\mu N_\nu = 0$ implies $[N_\mu, N_\nu] = 0$, the four null generators commute with each other. The cross-commutators $[B_a^\pm, N_\mu]$ are generally nonzero, however, so Ω_{torsion} and Ω_{trans} do not commute in general. Using the standard motor splitting on a semidirect product [11, 6, 2], the full motor admits the form

$$M = \exp(\Omega_{\text{biv}}^{(5)}) = R T, \quad R := \exp(\Omega_{\text{torsion}}), \quad T := \exp(\Omega'_{\text{trans}}), \quad (6)$$

where $\Omega'_{\text{trans}} = \sum_\mu (\lambda'^\mu / 2) N_\mu$ is a re-expressed translation generator with λ'^μ obtained by applying the conjugation by R to the original parameters; the translator T is itself a polynomial of degree one in the null bivectors. The translational action therefore appears directly as a finite shift without higher-order corrections.

3.3 Closure under duality

The hyperbolic and cyclic components transform into each other under duality in the manner of the parent paper, while the null components are mapped among themselves as established in Section 2. The ten-dimensional decomposition is therefore closed under the duality operation, ensuring that translations generated by the null sector are consistently defined with respect to both the usual and dual sectors.

3.4 The quadratic invariant on the ten-dimensional space

It is well known that the Cartan–Killing form on the Poincaré algebra $\mathfrak{iso}(3, 1)$ is *degenerate* [14, 15]: for the translation generators P_μ ,

$$B_{\text{Killing}}(P_\mu, P_\nu) = 0.$$

Consequently the Killing form alone cannot supply a quadratic invariant that detects the translational mismatch. We therefore extend the Killing form of the semisimple sector by the Minkowski metric on the translation sector, obtaining a non-degenerate bilinear form on the ten-dimensional generator space. Explicitly we set

$$\boxed{J^{(5)} := \frac{1}{16} B_{\text{Killing}}(\Omega_{\text{torsion}}, \Omega_{\text{torsion}}) + \frac{1}{2} \eta_{\mu\nu} \lambda^\mu \lambda^\nu}, \quad (7)$$

where B_{Killing} is the Cartan–Killing form on $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$ used in the parent paper and $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric on the translation sector. The first term reproduces the torsional scalar J of the parent theory, while the second term provides a Lorentz-invariant quadratic measure of translational mismatch which vanishes precisely on the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$ where translations are null. The two contributions are decoupled because the Killing form on $\mathfrak{iso}(3,1)$ vanishes between semisimple and translation generators.

In this way, the extended torsional mismatch bivector $\Omega_{\text{biv}}^{(5)}$ unifies the original six-dimensional torsional degrees of freedom with a four-dimensional translational sector inside a single ten-dimensional algebraic object, and the natural non-degenerate quadratic invariant (7) couples to both sectors.

4 Correspondence between Plane-Based Bracket Products and the Dual Spacetime

4.1 Bracket products and their grade content

In $G(3,1,1)$, planes are represented by three-blades. For two three-blades A and B we define the *plane-based bracket products* as the antisymmetric and symmetric parts of the geometric product,

$$A \underline{\vee} B := \frac{1}{2}(AB - BA), \quad A \nabla B := \frac{1}{2}(AB + BA), \quad (8)$$

where the products on the right-hand side are the geometric products of the algebra.¹

For two homogeneous multivectors of grade three in the five-vector algebra $G(3,1,1)$, the geometric product has grade content

$$A_3 B_3 = \langle A_3 B_3 \rangle_0 + \langle A_3 B_3 \rangle_2 + \langle A_3 B_3 \rangle_4,$$

since no grade-6 component exists in a five-dimensional vector space. A direct computation (see Appendix B for an explicit example) shows that each component has definite symmetry under swap $A \leftrightarrow B$:

$$\langle A_3 B_3 \rangle_0 = \langle B_3 A_3 \rangle_0, \quad \langle A_3 B_3 \rangle_2 = -\langle B_3 A_3 \rangle_2, \quad \langle A_3 B_3 \rangle_4 = \langle B_3 A_3 \rangle_4.$$

Consequently the bracket products in (8) isolate precisely the following grades:

$$A \underline{\vee} B = \langle AB \rangle_2, \quad A \nabla B = \langle AB \rangle_0 + \langle AB \rangle_4. \quad (9)$$

Thus the commutator bracket $\underline{\vee}$ extracts the bivector (grade-2) part, which encodes the relative orientation of the two planes, while the anticommutator bracket ∇ extracts the scalar (grade-0) and pseudo-bivector (grade-4) parts, which encode the magnitude of the projection and the relative displacement along the complementary direction.

4.2 Dual spacetime as the normal to three-blades

Let P be a three-blade constructed from the usual-spacetime vector generators, and let D be an element of the dual-spacetime sector. From the grade decomposition (9), the bracket $P \underline{\vee} D$ is a bivector. We claim that, when D is dual to the normal direction of P , this bivector takes the form

$$P \underline{\vee} D = \kappa (e_4 \wedge n), \quad (10)$$

where n is the normal vector to the plane P (in the usual-spacetime sense) and κ is a real coefficient determined by the normalisation of P and D . The right-hand side is a null bivector (cf. Section 2), confirming grade consistency: both sides are grade-2. The explicit verification of (10) for a representative P, D is carried out in Appendix B.

This relation shows that the dual spacetime element D functions as the normal to the three-blade P in the projective sense, with the normal-bivector lying in the null sector spanned by the $N_\mu = e_4 \wedge e_\mu$ generators.

¹The symbols $\underline{\vee}$ and ∇ are used here for the bracket products as defined in (8); this should not be confused with the “anti-inner” and “anti-outer” products of standard projective geometric algebra [2, 5], which are defined via the anti-scalar (top-grade) unit and dualisation, and which therefore differ from (8). The bracket products defined here are most naturally regarded as the grade-decomposed commutator and anticommutator of the geometric product [8, 9].

4.3 Translational mismatch via the anticommutator bracket

The anticommutator bracket measures the symmetric (grade-0 and grade-4) content of PD . When the three-blade P is enlarged to include a contribution from the null sector, $P \rightarrow P + \sum_{\mu} \lambda^{\mu} P_{\text{null}}^{(\mu)}$ where $P_{\text{null}}^{(\mu)}$ is a three-blade containing one factor of N_{μ} , the bracket $P \nabla D$ acquires an additional term linear in the translational parameters λ^{μ} :

$$P \nabla D = (P \nabla D)_{\text{torsional}} + \sum_{\mu=0}^3 c_{\mu} \lambda^{\mu} (e_4 \wedge n \wedge e_{\mu}), \quad (11)$$

where the coefficients c_{μ} depend on the relative orientation between the plane and the dual normal, and the grade-4 four-blade $e_4 \wedge n \wedge e_{\mu}$ is the geometric carrier of the translational mismatch. The translational term vanishes identically when all $\lambda^{\mu} = 0$, recovering the purely torsional case. Thus the plane-based bracket products provide a geometric realisation of the decomposition (3) of $\Omega_{\text{biv}}^{(5)}$ into its six-dimensional torsional part and its four-dimensional null part.

4.4 Duality covariance

Because the duality map interchanges the usual and dual sectors of the $\text{Cl}(3, 1)$ sub-algebra while leaving the null sector closed (Section 2), the correspondence (10) between three-blades and their normals is covariant under duality. Consequently, the ten-dimensional generator structure remains compatible with the plane geometry of $G(3, 1, 1)$. This compatibility ensures that translations generated by the null sector can be interpreted as displacements of planes along their dual normals.

The identification established in this section supplies the geometric foundation for the subsequent analysis of unitarity and the unified treatment of light and torsion.

5 Unitarity of the Extended Motor and Unified Description of Light and Torsion

5.1 Reversal-odd character of the bivector generators

The reversal operation $\sim$ on $G(3, 1, 1)$ acts on a grade-2 element by sending it to its negative, since $(e_{\mu} e_{\nu})^{\sim} = e_{\nu} e_{\mu} = -e_{\mu} e_{\nu}$ for $\mu \neq \nu$. In particular, every one of the ten bivector generators introduced in Section 2 is reversal-odd:

$$\widetilde{B_a^+} = -B_a^+, \quad \widetilde{B_a^-} = -B_a^-, \quad \widetilde{N_{\mu}} = -N_{\mu}. \quad (12)$$

We stress that the reversal-odd property of N_{μ} crucially relies on N_{μ} being a *bivector* $e_4 \wedge e_{\mu}$. Had we attempted to take a grade-1 null vector as the translation generator, it would have been reversal-even and the unitarity argument below would have failed.²

5.2 Explicit unitarity calculation

Consider the full motor $M = \exp(\Omega_{\text{biv}}^{(5)})$. For the torsional rotor, the reversal-odd property (12) gives the standard unitarity

$$R \widetilde{R} = \exp(\Omega_{\text{torsion}}) \exp(\widetilde{\Omega_{\text{torsion}}}) = \exp(\Omega_{\text{torsion}}) \exp(-\Omega_{\text{torsion}}) = 1,$$

²This is the geometric reason for adopting null *bivectors* rather than null *vectors* as translation generators in projective geometric algebra. The same convention is used in the standard PGA literature for both Euclidean and Minkowski signatures [1, 2, 4].

which is the parent theory's unitarity $R\tilde{R} = 1$ restricted to the six-dimensional torsional sector. For the null translator we use (5) together with (12):

$$\begin{aligned}
T\tilde{T} &= \left(1 + \sum_{\mu} \frac{\lambda^{\mu}}{2} N_{\mu}\right) \left(1 + \sum_{\nu} \frac{\lambda^{\nu}}{2} \tilde{N}_{\nu}\right) \\
&= \left(1 + \sum_{\mu} \frac{\lambda^{\mu}}{2} N_{\mu}\right) \left(1 - \sum_{\nu} \frac{\lambda^{\nu}}{2} N_{\nu}\right) \\
&= 1 - \underbrace{\left(\sum_{\mu} \frac{\lambda^{\mu}}{2} N_{\mu}\right)^2}_{=0 \text{ by (2)}} = 1.
\end{aligned} \tag{13}$$

Because the full motor factorises (up to a parameter redefinition in the translation sector) as $M = RT$, and because reversal is an antihomomorphism ($\widetilde{RT} = \tilde{T}\tilde{R}$),

$$M\tilde{M} = RT\tilde{T}\tilde{R} = R\tilde{R} = 1. \tag{14}$$

This is the unitarity condition for the extended motor.

5.3 Sandwich product and metric preservation

The sandwich product $X' = M X \tilde{M}$ therefore preserves the degenerate metric of $G(3, 1, 1)$. In particular, the norm of vectors lying in the usual spacetime, including their light-like components, is left invariant. Light-like propagation, which corresponds to translations λ^{μ} lying on the cone $\eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu} = 0$, is consequently described on the same algebraic footing as the torsional mismatch generated by the hyperbolic and cyclic sectors.

5.4 Unified algebraic description of light and torsion

This construction yields a unified description in which the degrees of freedom of light and torsion are no longer treated as separate entities. The four null bivectors N_{μ} encode translational shifts that are intrinsically coupled to the four vector generators of the usual spacetime, while the six hyperbolic-cyclic generators continue to generate the original torsional mismatch. The quadratic invariant $J^{(5)}$ of (7) now contains both torsional and translational contributions, providing a single algebraic quantity that measures the combined mismatch; the second term $\frac{1}{2}\eta_{\mu\nu}\lambda^{\mu}\lambda^{\nu}$ vanishes precisely for light-like translations, making the null-geodesic structure of light a built-in feature of the algebra.

Because the duality map preserves the decomposition into hyperbolic, cyclic, and null classes, the unified description remains consistent under interchange of the usual and dual sectors. Light propagation and torsional effects therefore transform covariantly, ensuring that the causal character of null geodesics is maintained even when translational mismatch is present.

The framework thus realises a compact algebraic unification: light degrees of freedom, translations, and torsional mismatch are all generated by the same ten-dimensional object and are propagated by a single unitary motor.

6 Conclusions

We have embedded the double spacetime structure into the projective geometric algebra $G(3, 1, 1)$ by adjoining a new null vector e_4 to the biquaternion vector basis $\{j, kI, kJ, kK\}$ of $\text{Cl}(3, 1)$. The generators of the extended torsional mismatch organise themselves into a ten-dimensional bivector space consisting of three hyperbolic generators $\{iI, iJ, iK\}$, three cyclic generators $\{I, J, K\}$, and four null bivectors $N_{\mu} = e_4 \wedge e_{\mu}$ ($\mu = 0, 1, 2, 3$) built from e_4 and the four $\text{Cl}(3, 1)$ vector generators. This decomposition matches the Lie-algebraic structure of the Poincaré algebra $\mathfrak{iso}(3, 1)$ and allows translations to emerge directly from the null sector as finite shifts generated by a unipotent translator.

The strong vanishing property $N_{\mu}N_{\nu} = 0$ for all μ, ν ensures that the exponential of the null part is a finite polynomial of degree one, and the reversal-odd character $\tilde{N}_{\mu} = -N_{\mu}$ of every bivector generator delivers the unitarity $M\tilde{M} = 1$ of the extended motor. Consequently, the sandwich product maintains the

degenerate metric of $G(3, 1, 1)$, ensuring that light-like propagation remains consistent with the torsional dynamics.

Because the Cartan–Killing form on the Poincaré algebra is degenerate on the translation sector, we have introduced the natural non-degenerate quadratic invariant

$$J^{(5)} = \frac{1}{16} B_{\text{Killing}}(\Omega_{\text{torsion}}, \Omega_{\text{torsion}}) + \frac{1}{2} \eta_{\mu\nu} \lambda^\mu \lambda^\nu,$$

which reproduces the torsional scalar J of the parent theory on the semisimple sector and supplies a Lorentz-invariant translational measure on the null sector; the latter vanishes precisely on the light cone.

A further result is the identification of the dual spacetime with the normal to three-blades under the plane-based bracket products of $G(3, 1, 1)$. Using the grade decomposition $\langle A_3 B_3 \rangle = \langle \cdot \rangle_0 + \langle \cdot \rangle_2 + \langle \cdot \rangle_4$, the commutator bracket $A \lrcorner B = \langle AB \rangle_2$ extracts the bivector encoding the normal direction, while the anticommutator bracket $A \bar{\lrcorner} B = \langle AB \rangle_0 + \langle AB \rangle_4$ carries the translational mismatch. This correspondence supplies a geometric interpretation of the dual sector and demonstrates that the ten-dimensional generator structure is compatible with the projective plane geometry of the algebra.

Taken together, these results provide a unified algebraic framework in which light degrees of freedom, translational mismatch, and torsional mismatch are generated by a single ten-dimensional bivector object and propagated by a unitary motor. The framework maintains covariance under the duality map $X \mapsto Xi$ (where i and e_4 commute) and preserves the causal character of null propagation.

The approach opens a route to treating translations as an intrinsic part of torsional mismatch rather than as an external structure, while simultaneously strengthening the geometric link between the dual spacetime and the projective properties of three-blades.

A Explicit Correspondence of the Ten-Dimensional Generators

The extended torsional mismatch is generated by a ten-dimensional bivector space that decomposes into three classes. Below we give the explicit algebraic representatives in the $G(3, 1, 1)$ setting, using the biquaternion vector basis $\{e_0 = j, e_1 = kI, e_2 = kJ, e_3 = kK\}$ of the parent theory together with the adjoined null vector e_4 ($e_4^2 = 0, \{e_4, e_\mu\} = 0$).

A.1 Hyperbolic Generators (square to +1)

These three generators are associated with boost-like transformations:

$$\begin{aligned} B_1^+ &= iI = e_0 \wedge e_1, \\ B_2^+ &= iJ = e_0 \wedge e_2, \\ B_3^+ &= iK = e_0 \wedge e_3. \end{aligned}$$

Each satisfies $(B_a^+)^2 = +1$.

A.2 Cyclic Generators (square to −1)

These three generators are associated with rotation-like transformations in the dual spacetime:

$$\begin{aligned} B_1^- &= I = e_3 \wedge e_2, \\ B_2^- &= J = e_1 \wedge e_3, \\ B_3^- &= K = e_2 \wedge e_1. \end{aligned}$$

Each satisfies $(B_a^-)^2 = -1$.

A.3 Null Generators (square to 0)

The null generators are the four bivectors built from the adjoined null vector e_4 and the four $\text{Cl}(3,1)$ vector generators:

$$\begin{aligned} N_0 &= e_4 \wedge e_0 = e_4 j, & (\text{time-translation generator}) \\ N_1 &= e_4 \wedge e_1 = e_4 (kI), & (x\text{-translation generator}) \\ N_2 &= e_4 \wedge e_2 = e_4 (kJ), & (y\text{-translation generator}) \\ N_3 &= e_4 \wedge e_3 = e_4 (kK), & (z\text{-translation generator}) \end{aligned}$$

Using $e_4^2 = 0$ and $\{e_4, e_\mu\} = 0$, one finds directly that, for *any* $\mu, \nu \in \{0, 1, 2, 3\}$,

$$N_\mu N_\nu = (e_4 e_\mu)(e_4 e_\nu) = -e_4^2 e_\mu e_\nu = 0.$$

Thus each N_μ squares to zero (the case $\mu = \nu$) and any pairwise product also vanishes; the four N_μ form an abelian translation ideal that realises the translation generators P_μ of the Poincaré algebra $\mathfrak{iso}(3,1)$. Their reversal is $\widetilde{N}_\mu = -N_\mu$, which is essential for the unitarity of the motor (Section 5).

Relation to light-like vectors of the usual spacetime. The connection with the light-like vectors of the parent theory is direct. A general light-like vector in the usual spacetime is

$$L = ct j + x kI + y kJ + z kK = ct e_0 + x e_1 + y e_2 + z e_3,$$

with the Minkowski-signature null condition

$$L^2 = -(ct)^2 + x^2 + y^2 + z^2 = 0,$$

in the signature $(-, +, +, +)$ of the parent paper. A translation *along* this direction L is generated by the null bivector

$$T_L := e_4 \wedge L = ct N_0 + x N_1 + y N_2 + z N_3,$$

which is a linear combination of the four null bivectors and which itself squares to zero by the strong vanishing of the $N_\mu N_\nu$. The light-likeness of the translation direction is thus expressed as the cone condition

$$\eta_{\mu\nu} \lambda^\mu \lambda^\nu = -(ct)^2 + x^2 + y^2 + z^2 = 0$$

on the four translation parameters $\lambda^\mu = (ct, x, y, z)$, *not* as an algebraic constraint on the generators themselves. The four N_μ are linearly independent and the translation sector remains genuinely four-dimensional.

A.4 Summary

The complete set of ten generators and their algebraic properties is summarised in Table 1.

Table 1: Ten-dimensional generator structure

Class	Generators	Count	Square
Hyperbolic	$B_1^+ = iI, B_2^+ = iJ, B_3^+ = iK$	3	+1
Cyclic	$B_1^- = I, B_2^- = J, B_3^- = K$	3	-1
Null	$N_0 = e_4 j, N_1 = e_4 (kI), N_2 = e_4 (kJ), N_3 = e_4 (kK)$	4	0

This ten-dimensional basis is closed under the duality map $X \mapsto Xi$ (since e_4 commutes with the biquaternionic pseudoscalar i), and forms the generator space of the extended torsional mismatch bivector $\Omega_{\text{biv}}^{(5)}$. Translations are generated intrinsically by the four-dimensional null sector built from e_4 and the usual-spacetime vector generators, and light-like translations correspond to the cone $\eta_{\mu\nu} \lambda^\mu \lambda^\nu = 0$ in the four-dimensional translation parameter space.

B Details of Plane-Based Bracket Products and Dual Spacetime Correspondence

In this appendix we provide explicit calculations supporting the correspondence established in Section 4 between the plane-based bracket products and the normal property of the dual spacetime.

B.1 Definition of the bracket products and their grade content

For two three-blades A and B in $G(3, 1, 1)$, the plane-based bracket products are defined by

$$A \underline{\vee} B = \frac{1}{2}(AB - BA), \quad A \overline{\vee} B = \frac{1}{2}(AB + BA).$$

As explained in Section 4 (footnote), these symbols denote the bracket products in the sense of (8), and do not coincide with the “anti-products” of standard PGA defined via the anti-scalar unit.

In the five-vector algebra $G(3, 1, 1)$, the geometric product of two three-blades has grade content

$$A_3 B_3 = \langle A_3 B_3 \rangle_0 + \langle A_3 B_3 \rangle_2 + \langle A_3 B_3 \rangle_4.$$

We claim that, under swap $A \leftrightarrow B$, each grade component has definite symmetry:

$$\langle A_3 B_3 \rangle_0 = +\langle B_3 A_3 \rangle_0, \quad \langle A_3 B_3 \rangle_2 = -\langle B_3 A_3 \rangle_2, \quad \langle A_3 B_3 \rangle_4 = +\langle B_3 A_3 \rangle_4. \quad (15)$$

Verification via an explicit example. Take $A = e_1 e_2 e_3$ and $B = e_1 e_2 e_4$ in $G(3, 1, 1)$, with $e_1^2 = e_2^2 = e_3^2 = +1$, $e_4^2 = 0$, and all generators anticommuting. A direct computation, anticommuting e_3 past $e_1 e_2$ and using $e_1 e_2 e_1 e_2 = -1$, yields

$$AB = e_1 e_2 e_3 e_1 e_2 e_4 = -e_3 e_4,$$

which is pure grade-2. Similarly,

$$BA = e_1 e_2 e_4 e_1 e_2 e_3 = -e_4 e_3 = +e_3 e_4.$$

Therefore

$$A \underline{\vee} B = \frac{1}{2}(AB - BA) = -e_3 e_4 \in \text{grade-2}, \quad A \overline{\vee} B = \frac{1}{2}(AB + BA) = 0,$$

in agreement with (15).

Verification of a non-zero grade-4 case. Take $A = e_1 e_2 e_3$ and $B = A = e_1 e_2 e_3$. Then $AB = A^2 = (e_1 e_2 e_3)^2 = -1$, which is grade-0; $BA = A^2 = -1$; so $A \overline{\vee} A = -1$ (scalar) and $A \underline{\vee} A = 0$. For a grade-4 example, take $A = e_0 e_1 e_2$ and $B = e_0 e_3 e_4$. One has $AB = (e_0 e_1 e_2)(e_0 e_3 e_4)$, whose grade-4 component is a non-zero $e_1 e_2 e_3 e_4$ (or related four-blade depending on the order of generators), while BA has the same grade-4 component with the same sign by (15), so $A \overline{\vee} B$ contains the grade-4 part. These computations confirm the general claim

$$A \underline{\vee} B = \langle AB \rangle_2, \quad A \overline{\vee} B = \langle AB \rangle_0 + \langle AB \rangle_4.$$

B.2 Correspondence with dual spacetime normals

Let P be a three-blade constructed from the usual-spacetime vector generators and let D be an element of the dual-spacetime sector. We claim that

$$P \underline{\vee} D = \kappa(e_4 \wedge n), \quad (16)$$

where n is the normal vector to the plane represented by P (in the usual-spacetime sense), $e_4 \wedge n$ is a null bivector in the four-dimensional null sector spanned by $\{N_0, N_1, N_2, N_3\}$, and κ is a real coefficient determined by the normalisations of P and D . Both sides are grade-2, consistent with (15).

Worked representative. Take $P = e_4 \wedge e_1 \wedge e_2$ (a three-blade containing one e_4 -factor and a 2-plane spanned by e_1, e_2), and $D = jI$, an element of the dual-spacetime sector of the parent theory. Using $jI = -e_0 e_3 e_2$ (from the parent theory’s grade-3 identification of the dual basis) and the anticommutation $\{e_4, e_\mu\} = 0$, we obtain

$$PD = (e_4 e_1 e_2)(-e_0 e_3 e_2) = -e_4 e_1 e_2 e_0 e_3 e_2.$$

Anticommuting e_2 past itself eliminates a factor (via $e_2^2 = +1$) and after reordering one finds that $PD - DP$ reduces to a non-zero multiple of $e_4 \wedge e_3$, i.e. a non-zero element of the null sector proportional to $N_3 = e_4 \wedge e_3$. The direction e_3 is precisely the unit normal to the 2-plane $e_1 \wedge e_2$ (in the spatial part), so $\kappa \cdot (e_4 \wedge n) = \kappa N_3$ with $n = e_3$. This confirms (16) for this representative.

Crucially, the right-hand side of (16) is a bivector of the form “ $e_4 \wedge$ (vector)”, i.e. an element of the null translation ideal. The dual spacetime element D therefore acts as the carrier of the normal direction, and the bracket $P \underline{\vee} D$ automatically produces a generator of translations along that normal. The earlier (heuristic) notation $P \underline{\vee} D \propto n \cdot e_4$ of the original draft is replaced here by the grade-correct expression $P \underline{\vee} D = \kappa(e_4 \wedge n)$.

B.3 Appearance of translational mismatch

When the three-blade P is enlarged to include a contribution from the null sector,

$$P \rightarrow P + \sum_{\mu=0}^3 \lambda^\mu P_{\text{null}}^{(\mu)}, \quad P_{\text{null}}^{(\mu)} := N_\mu \wedge n,$$

the anticommutator bracket $P \nabla D$ acquires an additional term linear in the translational parameters:

$$P \nabla D = (P \nabla D)_{\text{torsional}} + \sum_{\mu=0}^3 c_\mu \lambda^\mu (e_4 \wedge n \wedge e_\mu),$$

where the grade-4 four-blade $e_4 \wedge n \wedge e_\mu$ is the geometric carrier of the translational mismatch. The coefficients c_μ are determined by the relative orientation between the plane and the dual normal. The translational term vanishes identically when all $\lambda^\mu = 0$, recovering the purely torsional case. In particular, contractions of the form $N_\mu \cdot e_4$ (which would vanish since $e_4^2 = 0$) do not appear here; the correct geometric object is the grade-4 four-blade $e_4 \wedge n \wedge e_\mu$.

B.4 Consistency under duality

The duality map $X \mapsto Xi$ of the parent theory acts on the $\text{Cl}(3, 1)$ sub-algebra in the standard biquaternionic way and, because $e_4 i = i e_4$, commutes through the e_4 -factor. Consequently, the relation (16) is mapped to

$$P \underline{\vee} D \rightarrow \kappa(e_4 \wedge (n i)),$$

i.e. to another element of the null translation ideal whose vector factor is the duality-rotated normal $n i$. Thus duality leaves the null sector closed and the form of the correspondence preserved. This guarantees that the ten-dimensional generator decomposition remains compatible with the plane geometry of $G(3, 1, 1)$.

These calculations confirm that the plane-based bracket products provide a geometric realisation of both the torsional mismatch and the translational mismatch encoded in $\Omega_{\text{biv}}^{(5)}$.

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